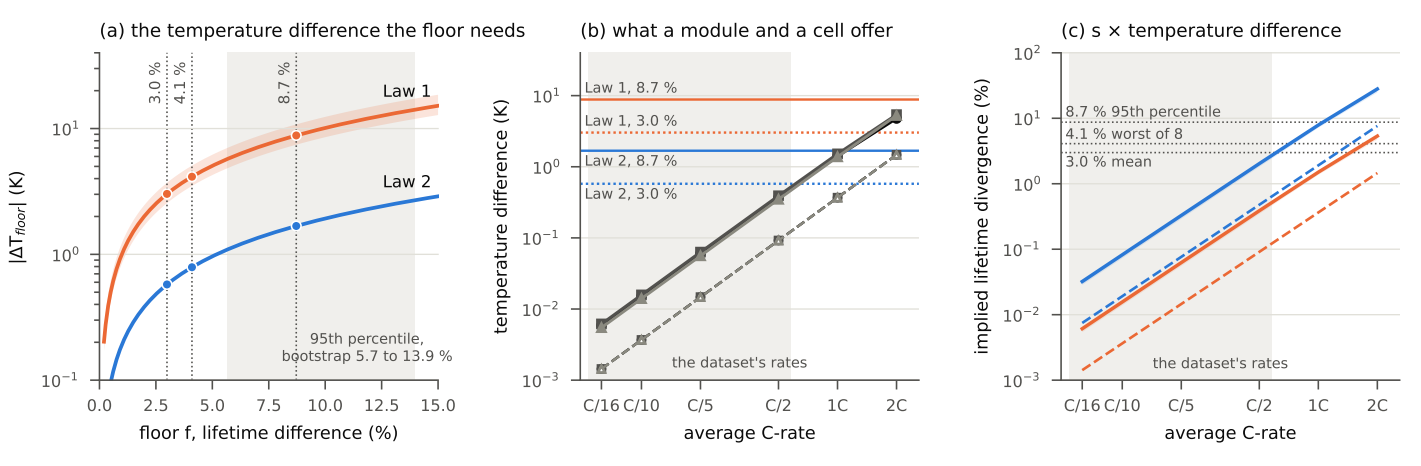
(a) the temperature difference the floor needs
3.0 %
4.1 %
8.7 %
Law 1
Law 2
95th percentile,
bootstrap 5.7 to 13.9 %
|ΔT_floor| (K)
floor f, lifetime difference (%)
(b) what a module and a cell offer
Law 1, 8.7 %
Law 1, 3.0 %
Law 2, 8.7 %
Law 2, 3.0 %
the dataset's rates
temperature difference (K)
C/16 C/10 C/5 C/2 1C 2C
average C-rate
(c) s × temperature difference
8.7 % 95th percentile
4.1 % worst of 8
3.0 % mean
the dataset's rates
implied lifetime divergence (%)
C/16 C/10 C/5 C/2 1C 2C
average C-rate
Law 1: NCA+LNO / Si-graphite 21700, C/2 charge
Law 2: LiFePO4 / graphite 26650, full model
eight-cell module, lumped
one cell, resolved
h = 5 W/(m² K)
h = 10 W/(m² K)
h = 25 W/(m² K)
bands: (a) Law 1 assumed noise, Law 2 C/16 to C/2; (c) h = 5 to 25